

400

345 456 70 60 Slide
Ruler
400px
(Log)
16 (John Napier 1550 – 1617) 20 “ ”

“10,000×100” “10,000” “100” “0” “1,000,000” , “10,000” ”10” , “100” “10”
“10,000×100”
“ ” “ $a^m \times a^n = a^{m+n}$ ” $10,000 \times 100 = 10^4 \times 10^2 = 10^{4+2} = 10^6 = 1,000,000$

16 20 8 7
18°30 0.3173047 0.111478920
550px|

16 A B

$$\cos \theta = \cos \varphi_A \cos \varphi_B \cos(\lambda_A - \lambda_B) + \sin \varphi_A \sin \varphi_B$$

	A B	A B		
A:	39°54 20	116°25 29	39.905556	116.424722
B:	48.8583	2.29451		

$$\begin{aligned}
\cos \theta &= \cos \varphi_A \cos \varphi_B \cos(\lambda_A - \lambda_B) + \sin \varphi_A \sin \varphi_B \\
&= \cos 39.905556^\circ \times \cos 48.8583^\circ \times \cos(116.424722^\circ - 2.29451^\circ) \\
&\quad + \sin 39.905556^\circ \times \sin 48.8583^\circ \\
&= 0.767103 \times 0.657923 \times \\
&\quad -0.40881 + 0.64152 \times 0.758084 \\
&= 0.28 \\
\theta &= 1.287 \\
(\text{rad}) \text{ AB} &= 6,378KM \times 1.287 = 8,208KM \quad 8,217KM
\end{aligned}$$

350

20

Robert Fritz “60 Boston Conservatory”

/ (Reference) Robert Fritz Problem Solving /

Spirit of Creating

(1973 40 144) 5

“ ”

_____ 2002 _____

(1770 – 1827)

form)——
16 4 45
“ ”
1805
(Minuetto) (Sonata
3 4 10

1824
= = = =
2025 4 Quatuor Ebene B (Op 130) (Große Fuge Op.
133) “ 3 ”

J. S. Bach

5

31

6

4

4

16

25 45

Fritz

600px

(VISION) (ACTION)
“ ”

50% /

(Apple)

iPhone iPad iTune (Steve Jobs) Steve

Jobs

= = =

80 MP3

(Hub)

CD

:

5G

Jony Ive “ ”

350px

iPod “ ”

400px|
399 iPod 2002
== == == ==
2001 2014 4 iPod 2015 50,000,000 iPod 2006 iPhone iPod 40% 2002 20
700px
700px

iTunes

iPod Mac iTunes 2000 iPod iTunes 2002 9%
2002
2003 4 iTunes store 200,000 6 100 6 100 2003 12 2500
2004 7 2003 4 2005 11 Walmart
2010 2
streaming

400 20 2000 3 iPod iTunes Digital
Hub iPod 2000
20 3 200
====
1.
2. 7
3. 3 2
1
4.
600px

1985 9 700 (workstation) NeXT SUN DEC Ma
90
1986 1987 1988 3 Mac 360 1984

1990
500
1993 600
1996
200
NextStep 4300
Ross Perot 200
NeXT
2000
NextStep 15
NeXT

1986
Kay
Story
Story
1986
Pixar
15
Pixar
Pixar
LucasFilm
5
1988
2006
George Lucas
1991
Pixar
Toy Story
74
Mac
Pixar
1995
3D
Toy
Alan
5
90%
“
”

iTunes iPod NeXT Pixar

- /
-

60
NeXT Pixar
“Fail Fast”

(Focus)

2010
NeXT Pixar
“
”
JIT

A

Pixar

Sywa Sung
3
“Pixar
2
3
Blackman
Delgado
(Star Trek)
Brad Bird
”
Pixar

A 1+1>2
 1984 Macintosh IBM IBM PC Burrell
 Smith Andy Hertzfeld Mac GUI Bill Atkinson Graphics Quick-
 Draw HyperCard Joanna Hoffman Bruce Horn Resource Manager
 Pixar (A Players)

600px

NeXT

iPod/iTunes(2001) iPhone(2007) iPad(2010)

Marsalis “ ” iTunes
 Marsalis “ ”
 Marsalis “ ”
 Jony Ive

300px

Ive

Walkman
 iPod 30

1983 ~ 84 28 Mackintosh

= = =

Pixar (Toy Story)

John Lasseter Hollywood
Pixar

Lasseter “ ”

Lasseter
1986 2 3D
“ 2 ”

“ 2 ”Lasseter Pixar
NeXT Lasseter 1986 8 (SIGGRAPH)

Pixar 10 Mackintosh

1988 Lasseter 30
Story) Pixar

Tin Toy 1888
Pixar

Lasseter Pixar 12.5% Pixar P

1991 (Woody)
(Toy Story)

1993 11

10%

1995 Pixar Pixar Toy Story IPO

Pixar 5 “

References

1. Fritz, Robert. *The path of least resistance for Managers.* (Berrett - Koehler Publishers, 1999)
2. Isaacson, Walter. *Steve Jobs.* (Simon and Schuster Paperbacks , 2011)

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